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Reduction – CO_2 is reduced into Carbohydrates (gain of electrons).

In some bacteria, oxygen is not evolved and is called as **non-oxygenic** and **anaerobic photosynthesis**. Examples: Green sulphur, Purple sulphur and green filamentous bacteria.

13.2.2 Significance of Photosynthesis

1. Photosynthetic organisms provide food for all living organisms on earth either directly or indirectly.
2. It is the only natural process that liberates oxygen in the atmosphere and balances the oxygen level.
3. Photosynthesis balances the oxygen and carbon cycle in nature.
4. Fuels such as coal, petroleum and other fossil fuels are from preserved photosynthetic plants.
5. Photosynthetic organisms are the primary producers on which all consumers depend for energy.
6. Plants provide fodder, fibre, fire wood, timber, useful medicinal products and these sources come by the act of photosynthesis.

13.2.3 Site of Photosynthesis

Chloroplasts are the main site of photosynthesis and both energy yielding process (Light reaction) and fixation of carbon dioxide (Dark reaction)that takes place in chloroplast. It is a double wall membrane bounded organelle, discoid or lens shaped, 4–10 µm in diameter and 1–33 µm in thickness. The membrane is a unit membrane and space between them is 100 to 200 A°. A colloidal and proteinaceous matrix called stroma is present inside.

Bioluminescence is the production and emission of light by a living organism. Bioluminescence is rare in true plants.

A team of MIT engineers have created living bioluminescent lamps out of watercress plants with the goal of one day replacing conventional electrical lighting with the glowing greenery.

A sac like membranous system called **thylakoid** or **lamellae** is present in stroma and they are arranged one above the other forming a stack of coin like structure called **granum** (plural grana). Each chloroplast contains 40 to 80 grana and each granum consists of 5 to 30 thylakoids.

Thylakoids found in granum are called grana lamellae and in stroma are called stroma lamellae. Thylakoid disc size is 0.25 to 0.8 micron in diameter. A thinner lamella called Fret membrane connects grana. Pigment system I is located on outer thylakoid membrane facing stroma and Pigment system II is located on inner membrane facing lumen of thylakoid. Grana lamellae have both PS I and PS II whereas stroma lamellae have only PS I. Chloroplast contains 30–35 Proteins, 20–30% phospholipids, 5–10% chlorophyll, 4–5% Carotenoids, 70S ribosomes, circular DNA and starch grains. Inner surface of lamellar membrane consists of small spherical structure called as **Quantasomes**. Presence of 70S ribosome and DNA gives them status of semi-autonomy and proves endosymbiotic hypothesis which says chloroplast evolved from bacteria. Thylakoid contains pigment systems which produces ATP and NADPH + H^+ using